

## Introduction

**Research Question:**  
Does Higher Unhealthiness Relate to Higher Popularity?

**Why It Matters:**  
I wanted to explore why fast-food chains are so popular and whether that popularity is linked to unhealthy menu items. This also emphasizes that consumers should not choose foods based only on popularity, since highly popular could be harmful to health.

**Hypothesis**  
I hypothesize that chains with higher unhealthy scores will also have higher popularity (measured by sales and store count)because foods high in fat, sugar, and sodium are often more appealing to consumers.

**Data Sources**

- Fast-food nutrition dataset
- Top 50 fast-food chains sales dataset

## Methods

$$Health\ Score = \left(\frac{sugar \times 4}{calories}\right) + \left(\frac{sat\_fat \times 9}{calories}\right) + \left(\frac{trans\_fat \times 9}{calories}\right) + \left(\frac{sodium}{calories}\right)$$

$\frac{0.10}{\% \text{ of Calories from Sugar}}$

$\frac{0.10}{\% \text{ of Calories from Sat Fat}}$

$\frac{0.01}{\% \text{ of Calories from Trans Fat}}$

$\frac{1.0}{Sodium per Calorie}$

- Created a custom Health Score inspired by the NPM Calculator framework
- Based on sugar, saturated fat, trans fat, sodium, and calories
- The higher the score, the more unhealthy the item is

## Results

The results show that there is not a perfect direct relationship between unhealthiness and popularity, but many high-selling chains still have moderately high unhealthy scores. For example, McDonald’s had the highest sales by a large margin, despite not having the highest unhealthy score, suggesting that factors such as brand recognition and store availability also influence popularity. Overall, the findings suggest that unhealthy menu profiles may contribute to popularity, but they are not the only reason.

- Liminations :**
- Different restaurants had different numbers of menu items
  - Popularity measured only by sales/store count
  - Does not account for price, advertising, or demand.

- Future Work**
- Investigate the effect of advertising/ brand influence
  - Compare menu pricing, affordability, and consumer access
  - Explore how this data can help better understand food insecurity and food choice behavior

## References

Pedersen, U. T. Fast Food Nutrition Dataset. Kaggle. Available at: <https://www.kaggle.com/datasets/ulrikthygepedersen/fastfood-nutrition>

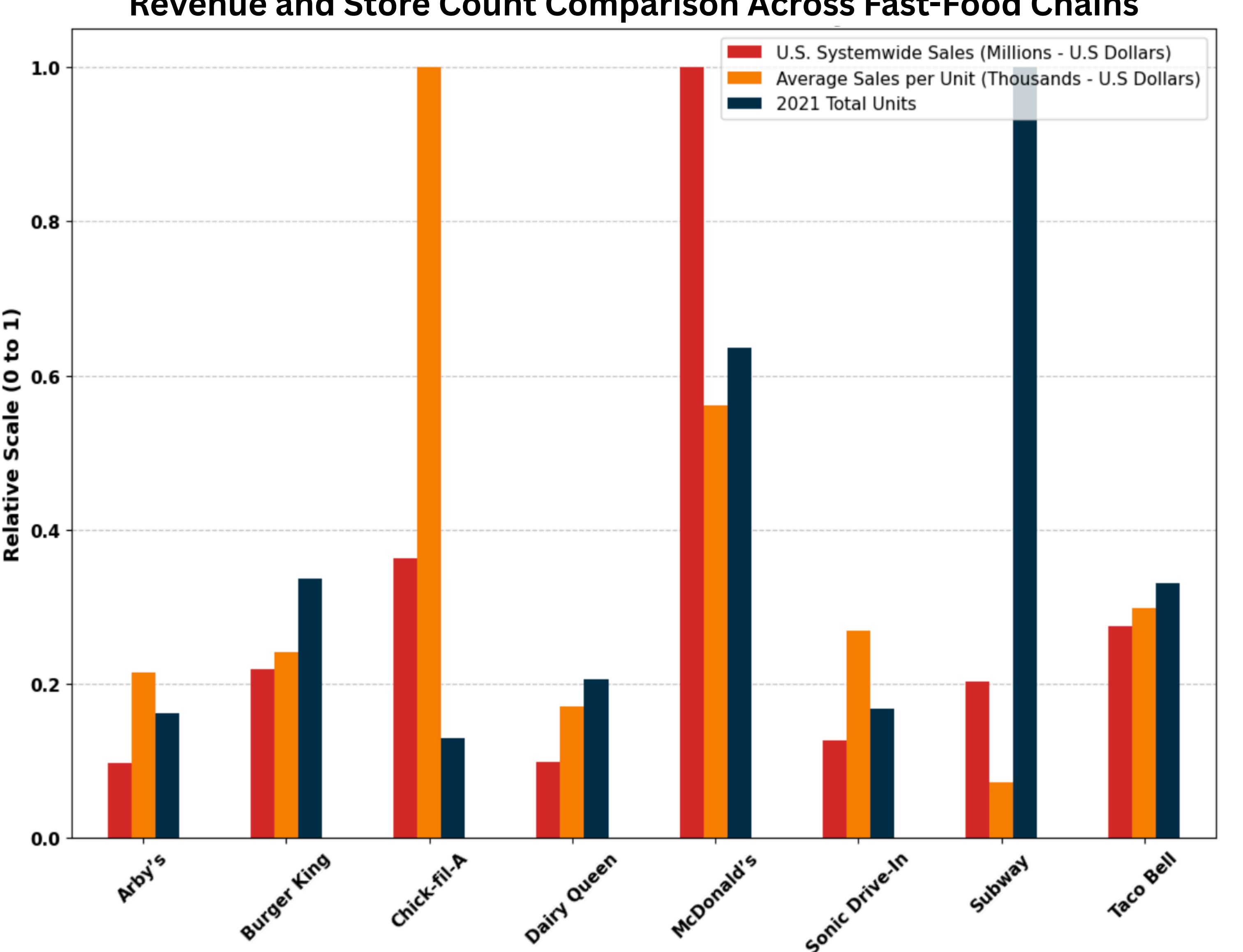
Stetsondone. Top 50 Fast Food Chains Dataset. Kaggle. Available at: <https://www.kaggle.com/datasets/stetsondone/top50fastfood>

Global Health Advocacy Incubator. Nutrient Profile Model (NPM) Calculator. Available at: <https://www.advocacyincubator.org/program-areas/food-and-nutrition/reducing-demand-for-unhealthy-food/npm-calculator>

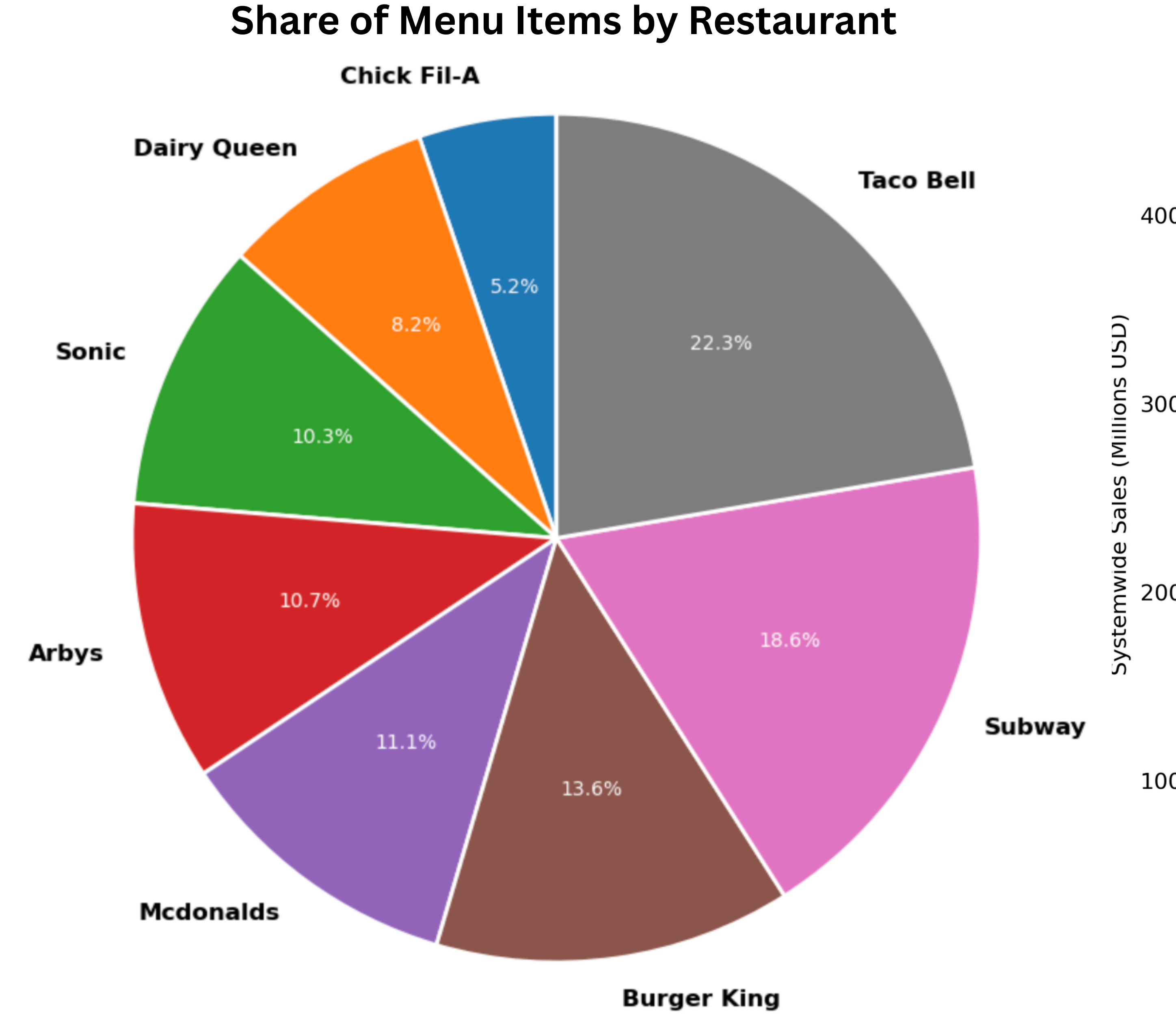
## Visualization

| Normalized Average Nutrition Values by Restaurant |          |        |         |           |       |
|---------------------------------------------------|----------|--------|---------|-----------|-------|
|                                                   | calories | sodium | sat_fat | trans_fat | sugar |
| restaurant                                        |          |        |         |           |       |
| Arbys                                             | 0.83     | 1.00   | 0.70    | 0.45      | 0.68  |
| Burger King                                       | 0.95     | 0.81   | 0.98    | 0.93      | 0.74  |
| Chick Fil-A                                       | 0.60     | 0.76   | 0.36    | 0.04      | 0.37  |
| Dairy Queen                                       | 0.81     | 0.78   | 0.91    | 0.73      | 0.57  |
| Mcdonalds                                         | 1.00     | 0.95   | 0.73    | 0.50      | 1.00  |
| Sonic                                             | 0.99     | 0.89   | 1.00    | 1.00      | 0.59  |
| Subway                                            | 0.79     | 0.84   | 0.54    | 0.23      | 0.91  |
| Taco Bell                                         | 0.69     | 0.67   | 0.58    | 0.27      | 0.33  |

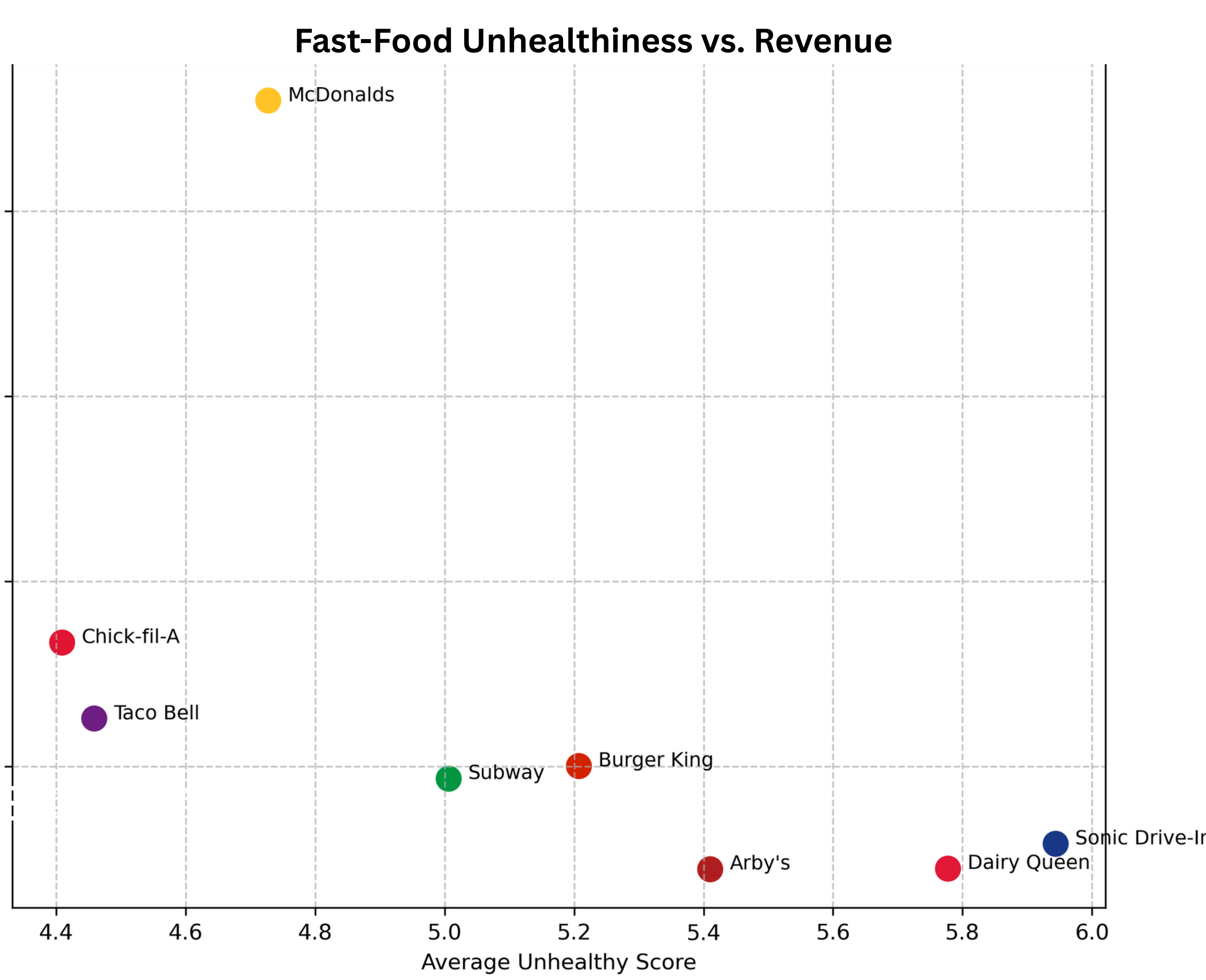
**Fig 1:** Shows normalized average nutrition values across fast-food chains, highlighting differences in calories, sodium, fat, and sugar



**Fig 2:** Compares normalized systemwide sales, average sales per unit, and total store count across fast-food chains. The figure highlights differences in overall popularity and scale, showing that some chains generate high revenue through strong per-store performance, while others rely on a larger number of locations.



**Fig 3 :** Shows the number of menu items analyzed for each chain, which may affect the stability of average unhealthy scores.



**Fig 4:** Compares each chain’s average unhealthy score with systemwide sales. Results suggest that popularity is influenced by more than nutrition alone.